



```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

int main()
{
    char *ptr1 = "name";
    char *ptr2;
    char *ptr3;
    char *ptr4;
    int i = 0;
    ptr2 = (char *)malloc(10);
    ptr3 = (char *)malloc(12);
    ptr4 = (char *)malloc(12);
    while (ptr2[i++] = ptr1[i]);
    printf("%s\n", ptr2);
    strcpy(ptr3, "1234");
    strcpy(ptr4, "ab0cdefghij");
}
```

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

int main()
{
    char *ptr1 = "name";
    char *ptr2;
    char *ptr3;
    char *ptr4;
    int i = 0;
    ptr2 = (char *)malloc(10);
    ptr3 = (char *)malloc(12);
    ptr4 = (char *)malloc(12);
    while (ptr2[i++] = ptr1[i]);
    printf("%s\n", ptr2);
    strcpy(ptr3, "1234");
    strcpy(ptr4, "ab0cdefghij");
}
```

QUESTION: 25

GROUP: Programming Basics SECTION: C Mark(s): 1

What will be the output of the following program?

TIME LEFT. 00:29:20

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <stdlib.h>
```

```
int main ()
```

```
char *ptr1 = "name" ;
```

```
char *ptr2;
```

```
char *ptr3;
```

```
char *ptr4;
```

```
int i = 0;
```

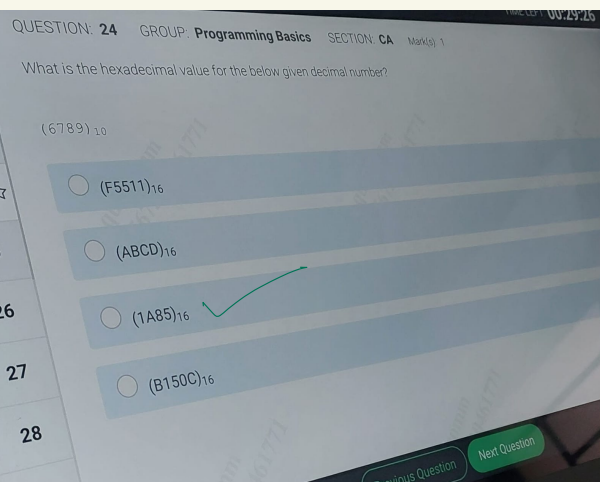
```
ptr2 = (char *)malloc (10) ;
```

```
ptr3 = (char *)malloc (12) ;
```

```
ptr4 = (char *)malloc (12) ;
```

Next Question

Previous Question



MIME LEF

00.29.26

QUESTION: 24

GROUP: Programming Basics

SECTION: CA

Mark(s) 1

What is the hexadecimal value for the below given decimal number?

(6789) 10

(F5511) 16

(ABCD) 16

26

(1A85)16

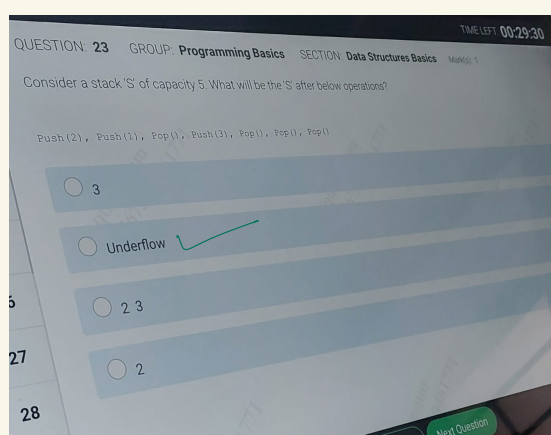
27

(B150C)16

28

Next Question

Dravions Question



TIME LEFT 00:29:30

QUESTION: 23 GROUP: Programming Basics

SECTION: Data Structures Basics Mark(s) 1

Consider a stack 'S' of capacity 5. What will be the 'S' after below operations?

Push (2), Push (1), Pop (), Push (3), Pop (), Pop (), Pop ()

3

Underflow ✓

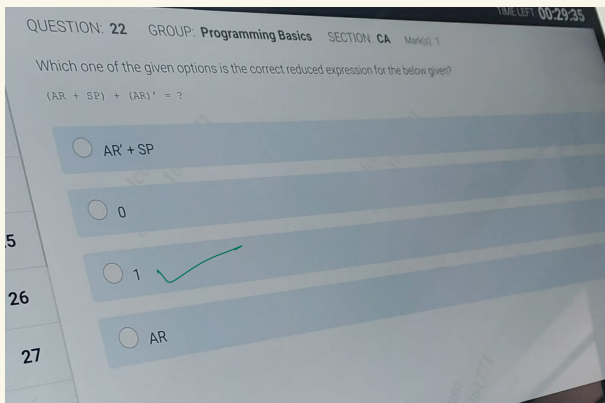
2 3

27

2

28

Next Question



TIME LEFT 00:29:35

QUESTION: 22

GROUP: Programming Basics SECTION: CA Mark(s) 1

Which one of the given options is the correct reduced expression for the below given?

$(AR + SP) + (AR)' = ?$

$AR' + SP$

0

5

1

26

AR

27

```

#include <stdio.h>
#include <stdlib.h>
#define MAXROW 3
#define MAXCOL 4

int main()
{
    int (*p) [MAXROW] [MAXCOL];
    p = (int (*) [MAXROW] [MAXCOL]) malloc (sizeof(*p));
    printf("%ld", sizeof(*p));
}

```

☐ 14

☒ 48

Previous Ques

Pro

2 Instructions

#include <stdio.h>

#include <stdlib.h>

#define MAXROW 3

#define MAXCOL 4

int main()

int (*p) [MAXROW] [MAXCOL] ;

p = (int (*) [MAXROW] [MAXCOL]) malloc (sizeof (*p)) ;

printf ("81d", sizeof (*p)) ;

27

14

28

48

29

Previous Ques

QUESTION: 59 GROUP: Machine Learning SECTION:

Model Selection Marks) 1

If we model by curve fitting of a straight line. Then, which one will be correct for the best fit.

Y

1

14

2

27

3

40

4

55

5

68

slope =13.6, intercept =0

57

slope =0, intercept = -13.6

58

slope =0, intercept =13.6

59

slope = 13.6, intercept =2

Next Question

Previous Question

60

QUESTION: 59 GROUP: Machine Learning SECTION: Model Selection Marks) 1

If we model by curve fitting of a straight line. Then, which one will be correct for the best fit.

X	Y
1	14
2	27
3	40
4	55
5	68

6

57

58

59 ☆

60

☒ slope =13.6, intercept =0

☐ slope =0, intercept = -13.6

☐ slope =0, intercept =13.6

☐ slope =13.6, intercept =2

QUESTION: 57 GROUP: Machine Learning

SECTION: Regularization and Bias

The substances that convey the impulse between neurons (via a synapse) are known as

MMELEFT 00:28:3

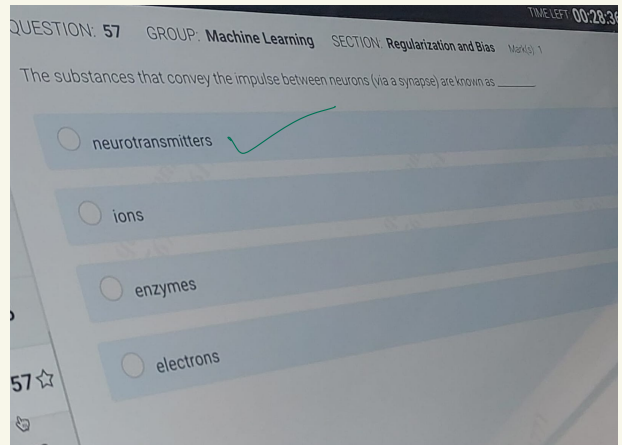
Mark(s). 1

neurotransmitters ✓

ions

enzymes

electrons



TIME LEFT. 00:28.34

QUESTION: 58 GROUP: Machine Learning

SECTION: Deep Learning Marks) 1

Which rule is used to perform classification for generating tree based classification algorithms?

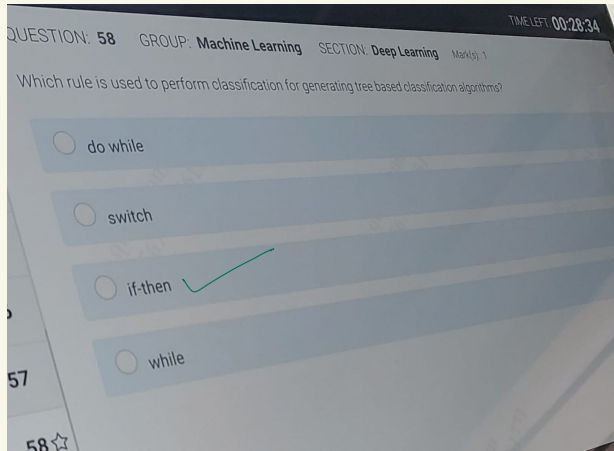
do while

switch

if-then

while

57



MaNp)

If a neuron consists of 3 inputs with values 4,5 and 6 and, weights 1,2 and 3 respectively. The associated transfer function is linear with a proportionality constant 2. Determine output for the neuron.

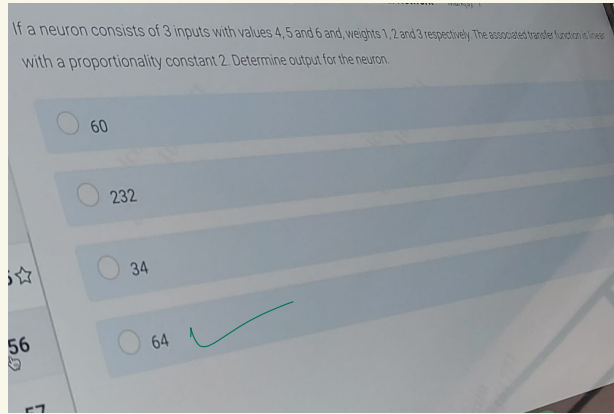
60

232

34

56

64



QUESTION: 56

GROUP: Machine Learning SECTION: Deep Learning Mark(e)

1

Decision tree can be used for:

I. Classification

II. Association

III. Regression

IV. Clustering

5

Only I and III

56

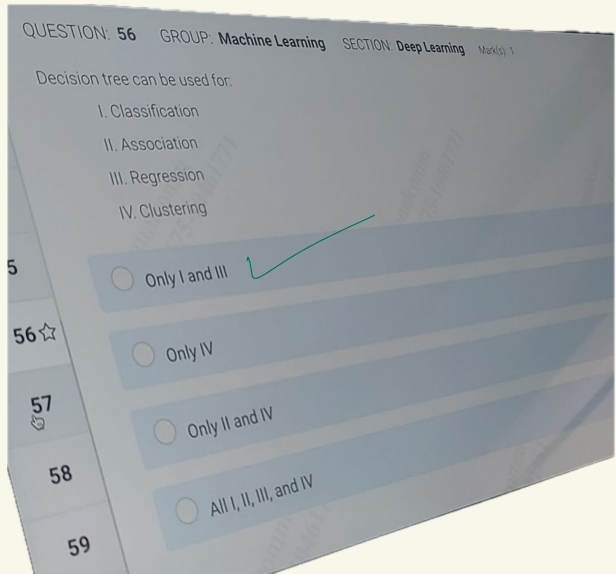
Only IN

57

Only II and IV

58

All I, II, III, and IV



QUESTION: 53 GROUP: Machine Learning SECTION: Graph Theory Mark(s): 1

If F be a bond of a connected graph G , then find $c(G - F)$.

☒ 2

☐ 3

☐ 6

☐ 4

QUESTION: 53 GROUP: Machine Learning

SECTION: Graph Theory Mark(s) 1

If F be a bond of a connected graph G , then find $c(G - F)$.

2

3

2

53¢

54

你

49

TIMELEFT: 00:28:49

The value of a correlation is reported by a researcher to be $r = -0.5$. then which of the following statement given in the answer option is correct?

50

51

52 T

53

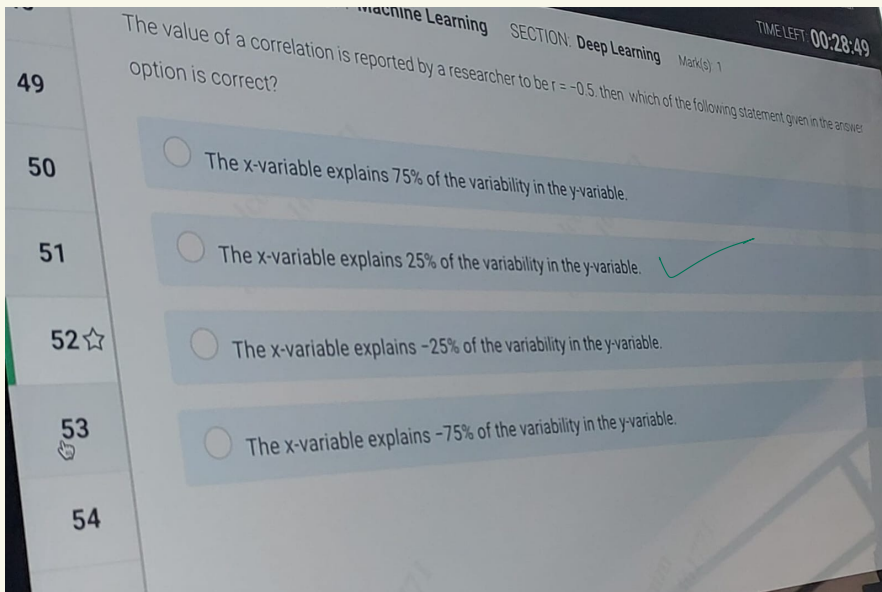
The x-variable explains 75% of the variability in the y-variable.

The x-variable explains 25% of the variability in the y-variable.

The x-variable explains -25% of the variability in the y-variable.

The x-variable explains -75% of the variability in the y-variable.

54



GROUP: Machine Learning SECTION: OOPs Mark(a) 1

Consider the statement given below:

"Software entities such as classes, modules, functions, etc. should be open for extension, but closed for modification"

Which one of the design principles given in the options expresses the above statement?

2

53

Dependency Inversion Principle (DIP)

54°

Liskov Substitution Principle (LSP)

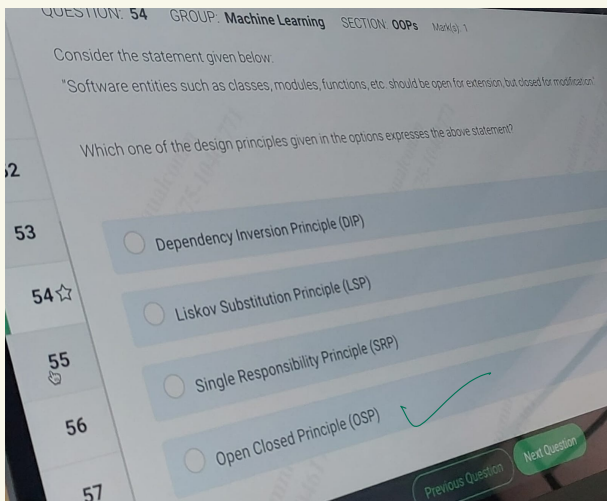
55

56

Single Responsibility Principle (SRP)

Open Closed Principle (OSP)

Next



TIME LEFT.

00:29:02

QUESTION: 49

GROUP: Machine Learning SECTION: Neural Network

Mark(s) 1

Consider a two class classification problem where the data set follow the XOR patten and can ft be clasified with Perceotor

No

Yes

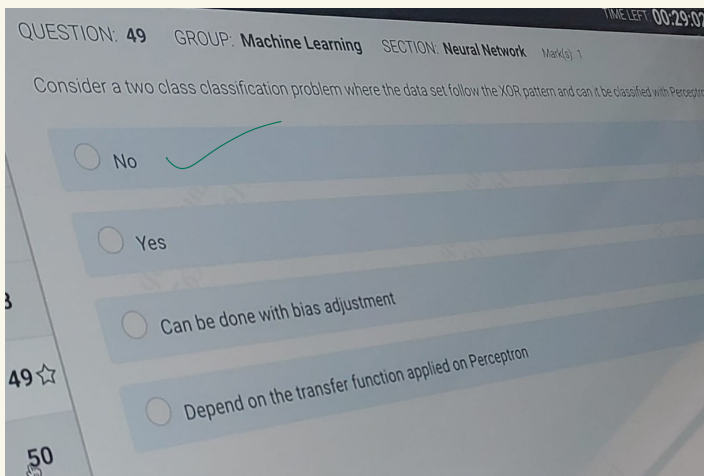
3

Can be done with bias adjustment

49

Depend on the transfer function applied on Perceptron

50



GROUP: Machine Learning SECTION:

Algorithm Marks). 1

What will be the return value on calling the function func(4,3)?

```
int funclint x, int y)
```

```
int count = 0;
```

```
x = x + 1;
```

```
if (x >= 0)
```

```
return x + y;
```

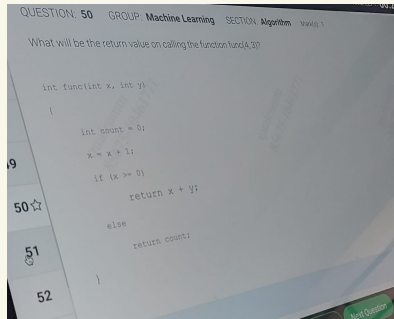
```
50°
```

```
else
```

```
return count;
```

```
52
```

Next



8

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QUESTION: 47

GROUP: Machine Learning SECTION: Regularization and Bias
Marks) 1

While minimizing augmented error function $E = E + \lambda \sum w_i^2$, what is the possible outcome with different values of λ ?

If λ is large, current instances of training data are used

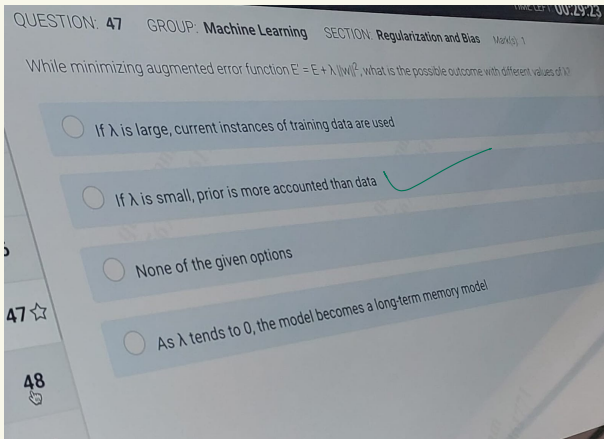
If λ is small, prior is more accounted than data

None of the given options

47 ÷

As λ tends to 0, the model becomes a long-term memory model

48



GROUP: Machine Learning SECTION: Text Mining Mark(s) 1
10gTotal Number documents/Number of Documents
containing term).

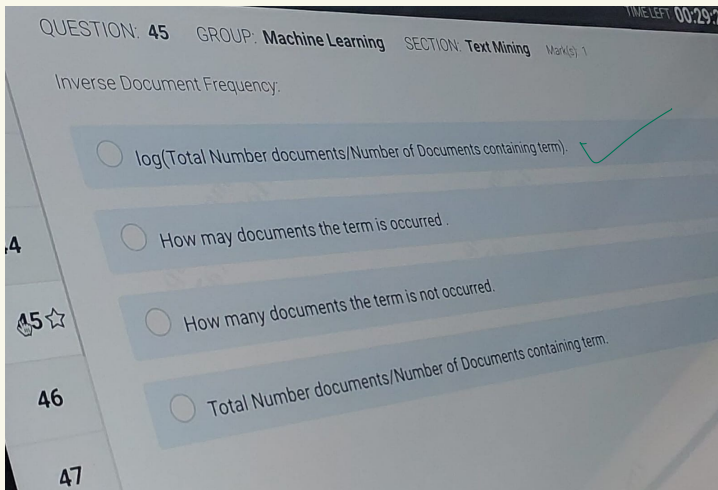
15

46

How may documents the term is occurred.

How many documents the term is not occurred.

Total Number documents/Number of Documents containing
term.



QUESTION: 48

GROUP: Machine Learning SECTION: Deep Learning Mark(s) 1

Identify the limitations of Naive Bayesian Classifier:

- I. Hard to implement
- II. Dependencies exist among variables
- III. Loss of accuracy
- IV. Class conditional independence

TIME LEFT: 00:29:05

7

Only II, III, and IV

48公

Only I, II, III, and IV

49

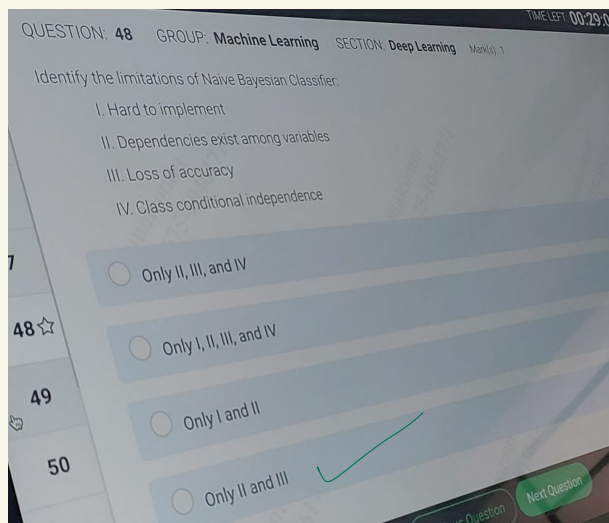
Only I and II

50

Only II and III

Next Question

alis Question



TIME LEFT 00:29.33

QUESTION: 43 GROUP: Machine Learning SECTION: SGD or Online learning Maria) 1

Which of the following functions are convex?

I) $\sin(x)$

46

II) $\min(f_1(x), f_2(x))$, where f_1 and f_2 are convex IV) $\max f_1(x), f_2(x)$, where f_1 and f_2 are convex

Both (I) and (II)

Both (II) and IV)

47

Both (I) and (III)

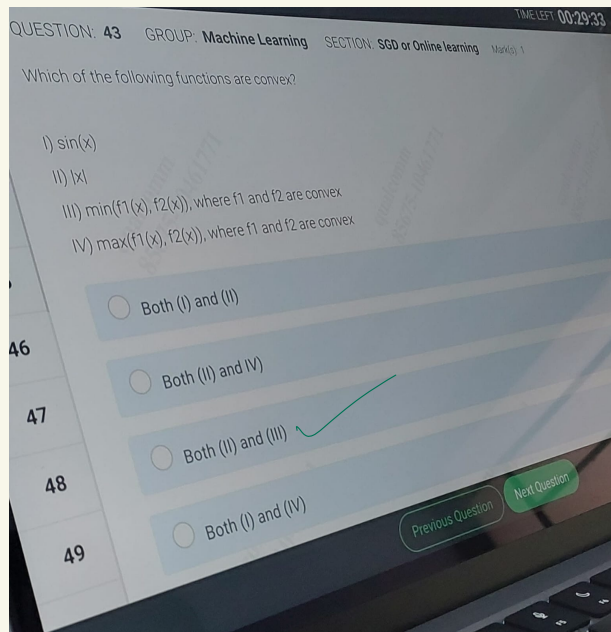
48

Next Question

Both (1) and (IV)

Previous Question

49



QUESTION: 44

GROUP: Machine Learning SECTION:

OOPs

Which of the below given option(s) is/
are the benefit (s) of an Inheritance?

TIME LEFT: 00:29:30

Mark(s), 1

I) Reusability of code

II) Code sharing

III) Consistency of interface

All (I),(II) and (II)

46

Only (II) and (II)

47

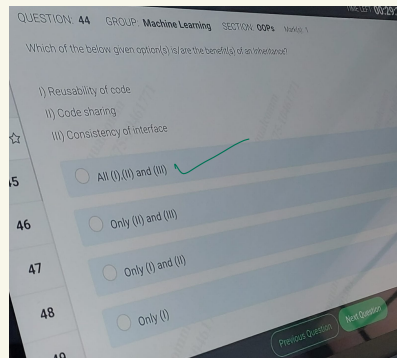
Only (1) and (II)

48

Only (I)

Next Question

Previous Question



QUESTION: 41

GROUP: Machine Learning SECTION: Recommender Systems

TIME LEFT: 00:29:41

Mark(s). 1

A Computer Vision was analyzing images traffic of in the city and identifying vehicles that are violating the rules, In this process. t was utilizing a deep learning algorithm and a neural network. Which of the following statements are valid with respect to the above context?

i) Deep learning algorithm enables computers to learn itself about the visual data
ii) Neural Network helps deep learning algorithm by breaking the images into pixels and provide appropriate tags to ther

Only i

Only ii

46

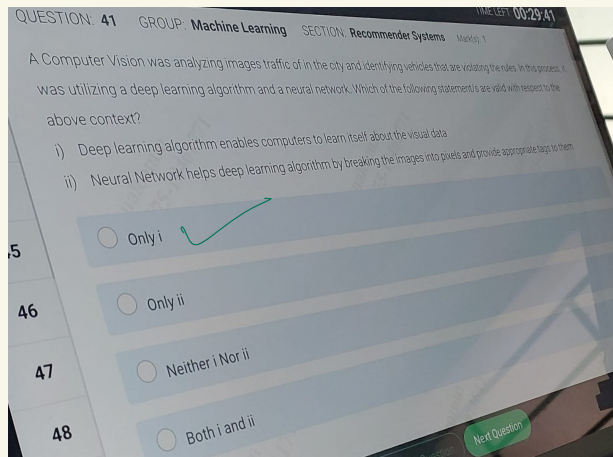
47

Neither i Nor ii

48

Both i and ii

Next Question



QUESTION: 42 GROUP: Machine Learning

SECTION: SGD or Online learning

Which among the following are not a convex loss function?

Mark(s) 1

Misclassification loss

Logistic loss

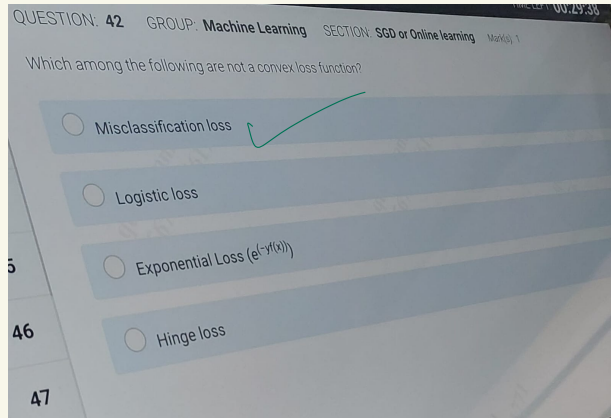
5

Exponential Loss ($e^{y f(x)}$)

46

Hinge loss

47



What will be the output of the program given below?

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
int m[4] = 15, 2, 4, 0);
```

```
int *p = m;
```

```
int t=0;
```

```
int i = 1;
```

```
while (t = *p++)
```

```
{
```

```
    (i++ <= 4) ? printf ("sd\n", t) :
```

```
    printf ("errorn") :
```

```
do
```

```
while
```

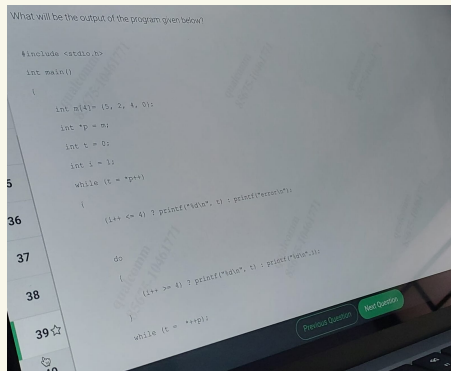
```
{
```

```
    (i++ >= 4) ? printf("&d\n", t) : printf ("&d\n",  
    i) :
```

```
Next Question
```

```
39 d
```

```
while (t = *++p) ;
```



5
2
4

QUESTION: 40

GROUP: Programming

Basics SECTION: CA

Mark(s) 1

Which one of the given options is the correct conversion value of the number (4ED)₁₆ in binary format?

TIME LEFT 00:21:53

1

1110 0100 1101

32

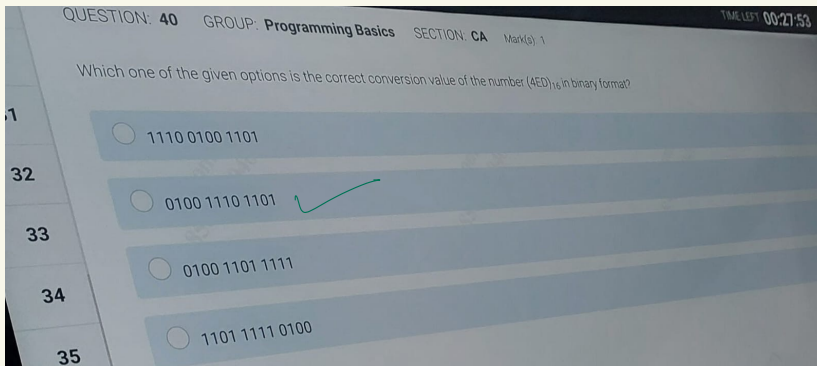
0100 1110 1101

33

0100 1101 1111

34

1101 1111 0100



MUEOTIUN: 37

GROUP: Programming Basics SECTION: Operating System Fundamentals Marks) 1

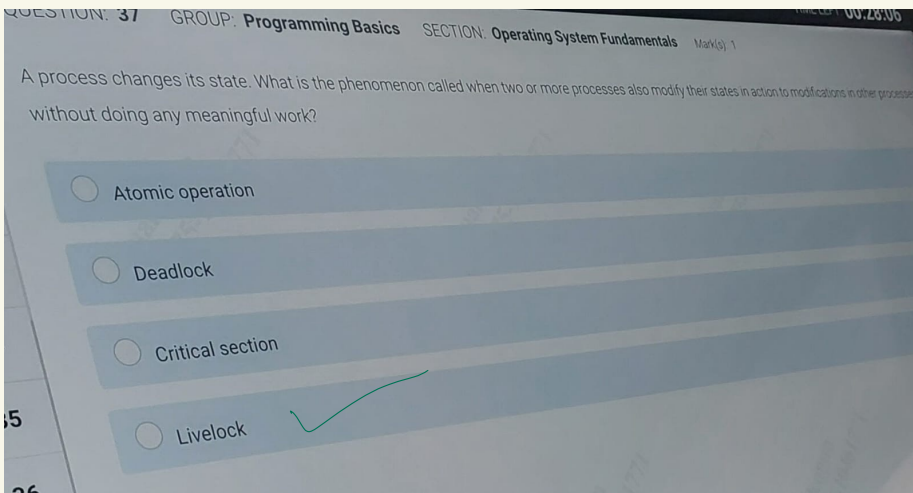
A process changes its state. What is the phenomenon called when two or more processes also modify their states in action to modifications in other processes without doing any meaningful work?

Atomic operation

Deadlock

Critical section

Livelock



allayud Untaval Chandidal

TIMELEFT 00:28:03

QUESTION: 38

GROUP: Programming Basics SECTION: CA Mark(s) 1

What is the decimal form for the below given hexadecimal number?

1CDE

9513

7787

7391

35

6596

36

QUESTION: 38 GROUP: Programming Basics SECTION: CA Mark(s) 1 TIMELEFT: 00:28:03

What is the decimal form for the below given hexadecimal number?

1CDE

☐ 9513

☐ 7787

☒ 7391

☐ 6596

35

36

QUESTION: 35

GROUP: Programming Basics SECTION: Operating System Fundamentals

Which method is used to handle negative synchronization in RTOS?

TIME LEFT 00:28:17

Mark(s). 1

Semaphores

Event Flags

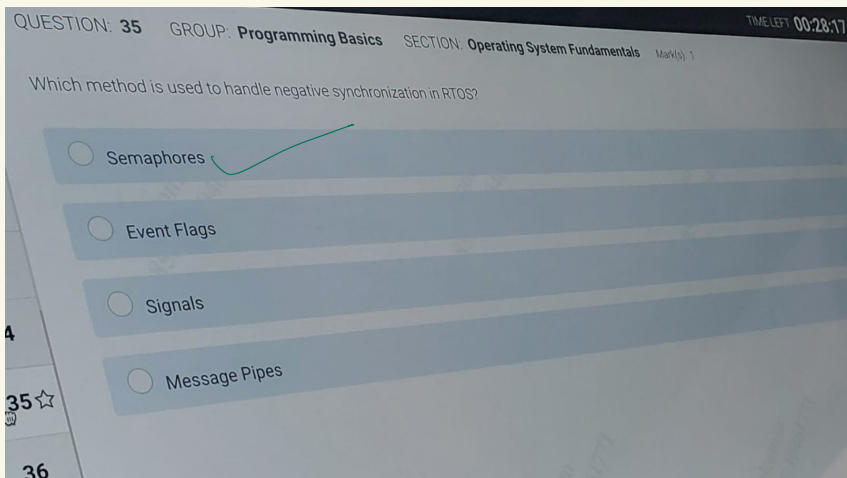
Signals

4

Message Pipes

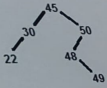
35

36



QUESTION: 36 GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

What is the total number of internal nodes in the below given binary tree?



☐ 2

☐ 3

☐ 6

☐ 4

QUESTION: 36

GROUP: Programming Basics

SECTION: Data Structures Basics

Mark(s) 1

What is the total number of internal nodes in the below given binary tree?

45

TIME LEFT 00:28:09

30

50

22

48

49

2

4

35

6

36

37

QUESTION: 33 GROUP: Programming Basics SECTION: CA

Mark(s) 1

Which of the below given is correct about logic gates?

TIME LEFT 00:28:28

I) TTL (Transistor-Transistor Logic) is a type of logic gate.

II) The logic state of a terminal can never change.

Only (I)

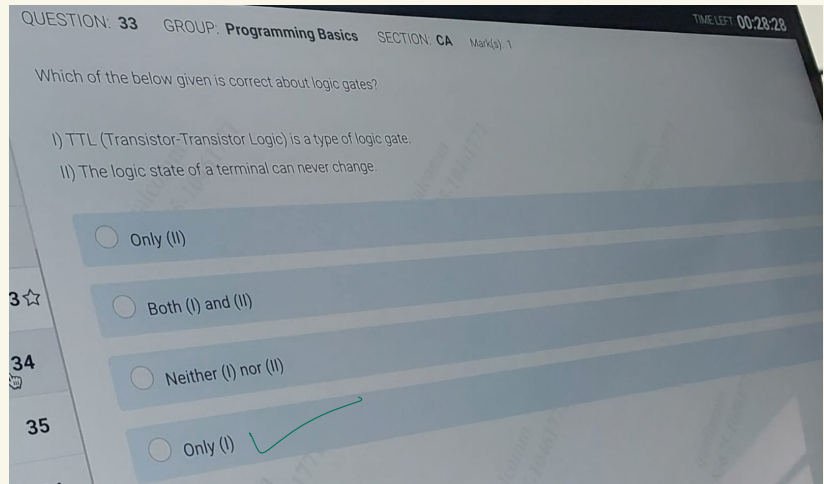
Both (I) and (II)

34

Neither (I) nor (II)

35

Only (I)



TIME LEFT 00:28:31

QUESTION: 32

GROUP: Programming Basics SECTION: Operating System Fundamentals Marks) 1

With an increase in time quantum when using Round Robin CPU scheduling algorithm, the round robin algorithm behaves like which of the algorithm given in options?

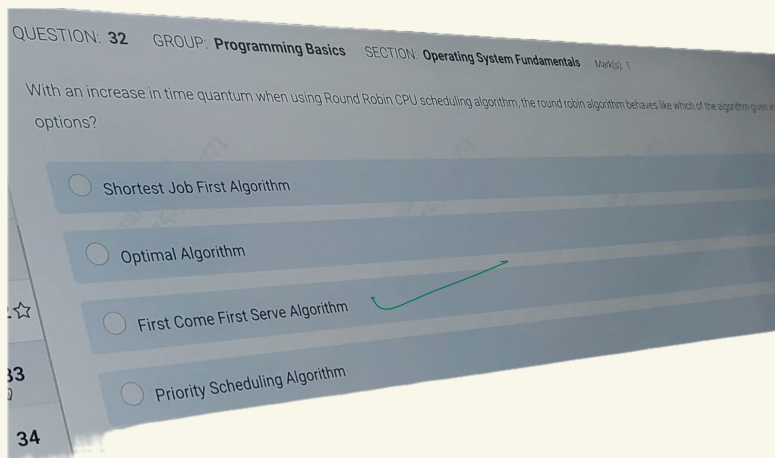
Shortest Job First Algorithm

Optimal Algorithm

First Come First Serve Algorithm

Priority Scheduling Algorithm

34



QUESTION: 34 GROUP: Programming Basics SECTION: Error Finding Mark(9) 1

Which of the line number from the given below program will give a compile time error.

Mier da Unaral Chandual

TIME LEFT 00:28:24

```
#include <stdio.h>
```

```
int a;
```

```
a = 30, 40;
```

```
int b = 1, 2;
```

```
//Line 3
```

```
//Line 4
```

```
int main ()
```

```
36
```

```
37
```

```
int c = 3, 4; //Line 8
```

```
int d;
```

```
d = 10, 20;
```

```
//Line 10
```

```
printf"%d &d %d
```

```
8d ", a, b, c, d);
```

```
return 0;
```

```
38
```

No error

Next Question

costion

QUESTION: 34 GROUP: Programming Basics SECTION: Error Finding Mark(9) 1

Which of the line number from the given below program will give a compile time error:

```
#include <stdio.h>
```

```
int a;
```

```
a = 30, 40; //Line 3
```

```
int b = 1, 2; //Line 4
```

```
int main()
```

```
{
```

```
int c = 3, 4; //Line 8
```

```
int d;
```

```
d = 10, 20; //Line 10
```

```
printf("%d %d %d %d ", a, b, c, d);
```

```
return 0;
```

```
}
```

No error

Which one of the given options will get searched in the given list by binary search technique in 3rd pass?

12 18 41 47 54 66 74 92 96
3 3 3 3 3

☐ 18

☐ 74

☒ 12

☐ 47

QUESTION: 31

GROUP: Programming Basics SECTION: Data Structures Basics Mark(s): 1

Which one of the given options will get searched in the given list by binary search technique in 3rd pass?

TIME LEFT 00:28:35

12 18 41 47 54 66 74 92 96

18

74

12

32

33

47